

A PROJECT TITLE ON

Customer Purchase Prediction Using Logistic Regression and Random Forest Machine Learning Algorithms

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Name of Title: Customer Purchase Prediction Using Logistic Regression and Random Forest Machine Learning Algorithms

ABSTRACT

Customer purchase prediction is an important application of machine learning that enables businesses to identify whether a customer is likely to purchase a product based on their demographic and behavioural information. Accurate prediction helps companies improve marketing strategies, enhance customer satisfaction, reduce advertising costs, and increase overall sales.

The objective of this project is to develop a predictive model using two supervised machine learning algorithms: Logistic Regression and Random Forest. The dataset used for this project contains customer information such as age, annual income, gender, and spending behaviour. Before training the models, the data undergoes preprocessing, including handling missing values, encoding categorical variables, feature scaling, and splitting into training and testing datasets.

Logistic Regression is used as a baseline classification model because of its simplicity, efficiency, and interpretability. Random Forest, an ensemble learning algorithm, is employed to improve prediction accuracy by combining multiple decision trees. Both models are trained using the prepared dataset and evaluated using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix. The results obtained from both algorithms are compared to determine the better-performing model for customer purchase prediction.

The project demonstrates how machine learning techniques can assist businesses in making data-driven decisions. Predicting customer purchasing behaviour enables organisations to identify potential buyers, personalise marketing campaigns, optimise resource allocation, and improve customer engagement. The comparison between Logistic Regression and Random Forest provides valuable insights into the strengths and limitations of both algorithms for binary classification problems. Overall, this project highlights the practical application of machine learning in customer analytics and business intelligence while providing a scalable solution for purchase prediction in real-world scenarios.

INTRODUCTION

Machine learning has become one of the most significant technologies in modern business analytics. Organisations generate large amounts of customer data every day, making it essential to analyse this information effectively. Predicting customer purchase behaviour helps businesses understand customer preferences and improve their decision-making processes.

This project focuses on predicting whether a customer will purchase a product using supervised machine learning techniques. Two classification algorithms, Logistic Regression and Random Forest, are implemented and compared based on their predictive performance.

The project begins with data preprocessing, including cleaning the dataset, encoding categorical variables, scaling numerical features, and dividing the data into training and testing sets. The models are then trained and evaluated using various performance metrics.

Logistic Regression provides a simple and interpretable statistical approach for binary classification, while Random Forest combines multiple decision trees to improve prediction accuracy and reduce overfitting. Comparing these models helps identify the most suitable algorithm for customer purchase prediction.

The outcome of this project demonstrates how machine learning can support businesses in targeted advertising, customer segmentation, personalised recommendations, and sales forecasting. It also provides practical experience in implementing machine learning algorithms using Python and popular libraries such as Pandas, NumPy, Scikit-learn, and Matplotlib.

SOFTWARE REQUIREMENTS

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB (8 GB recommended)
- Hard Disk: 20 GB free space
- Operating System: Windows 10/11, Linux, or macOS

Software Requirements

- Python 3.10 or above
- Visual Studio Code
- Git and GitHub

Python Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Machine Learning Algorithms Used

- Logistic Regression
- Random Forest Classifier

Dataset

- Customer Purchase Prediction Dataset (CSV)

Signature of the Guide

Course Instructor